# Selective Cerebral Perfusion Augmentation During Cardiac Arrest Using Dynamic Aortic Occlusion: A Pilot Study

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## BACKGROUND & OBJECTIVE

- Neurologic outcomes after prolonged pediatric in-hospital cardiac arrest remain poor.
- Even high-quality standard CPR generates only a fraction of normal cerebral blood flow.
- Physiology-directed CPR improves outcomes in large-animal models, but it can only do as much as compressions and vasopressors allow.
- Partial aortic occlusion (AO) redirects the blood flow CPR generates toward the heart and brain but risks proximal hyperperfusion and distal ischemia. Dynamically adjusting it to personalized cerebral optical measurements may mitigate these risks.

**OBJECTIVE**

**Test NOM-guided dynamic AO (dAO) vs. Depth-Guided CPR**

**Feasibility of real-time dynamic AO** | **↑ Time within cerebral $HbO_2$ target** | **↑ Coronary perfusion & cerebral oxygenation**

## EXPERIMENTAL DESIGN AND STUDY TIMELINE

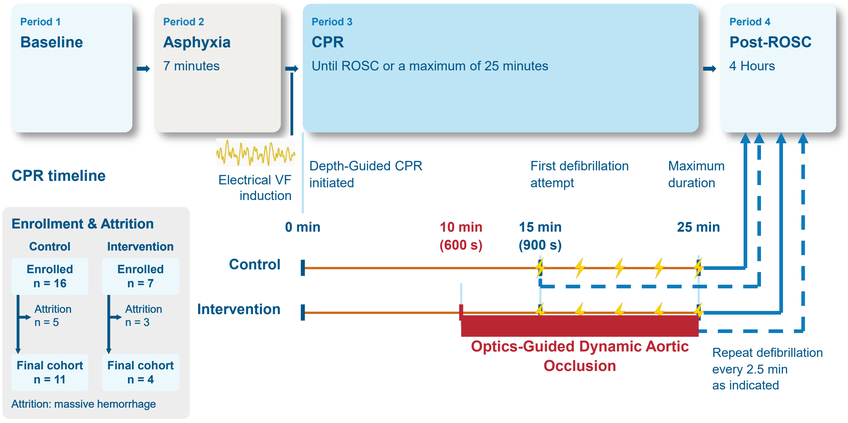



Figure 1. Experimental design and study timeline

## OUTCOMES

**1 PRIMARY OUTCOME**

**Time within the cerebral $HbO_2$ target range**

Before: **5-10 minutes of CPR** → After: **10-15 minutes of CPR**

% of CPR time with cerebral $HbO_2$ in the personalized target range

**2 SECONDARY OUTCOMES**

**Systemic and cerebral perfusion metrics**

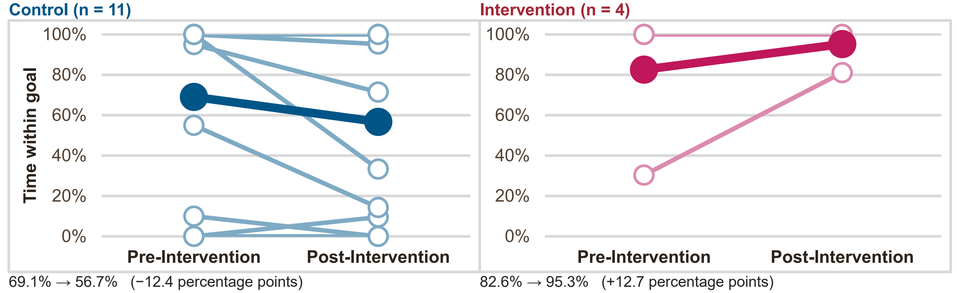
| | | | | |
|---|---|---|---|---|
| **Systemic** | Systolic BP | Diastolic BP | MAP | CoPP |
| **Cerebral oxygenation and perfusion** | $StO_2$ | rTHC | rDCS | |

**Expressed as:** difference-in-differences (intervention − control) and individual trajectories

## RESULTS

### Proportion of CPR Time Within Goal $HbO_2$

Goal ≥1.3 above 1 min Baseline | Pre: 5-10 min | Post: 10-15 min




69.1% → 56.7% (−12.4 percentage points)

82.6% → 95.3% (+12.7 percentage points)

**Difference-in-differences: +25.1 percentage points (95% CI −12.3 to +62.4)**

Post-window comparison: Welch p = 0.020

Figure 2. Proportion of CPR time within HbO2 goal, pre- vs post-intervention (Control n = 11, Intervention n = 4).

### Feasibility & Safety

**100%**

4/4 animals

**TECHNICAL SUCCESS**

Successful access, balloon placement, and dynamic optics-guided titration

**SAFETY EVENTS & COMPLICATIONS**

| | | | |
|---|---|---|---|
| **0** | Aortic rupture | **0** | Intracranial hemorrhage |
| **2/4** | Distal arterial thrombosis | **n = 3** | Right atrial injury from PAC (excluded from analysis cohort) |

### Individual physiologic trajectories during CPR: Control vs. Intervention

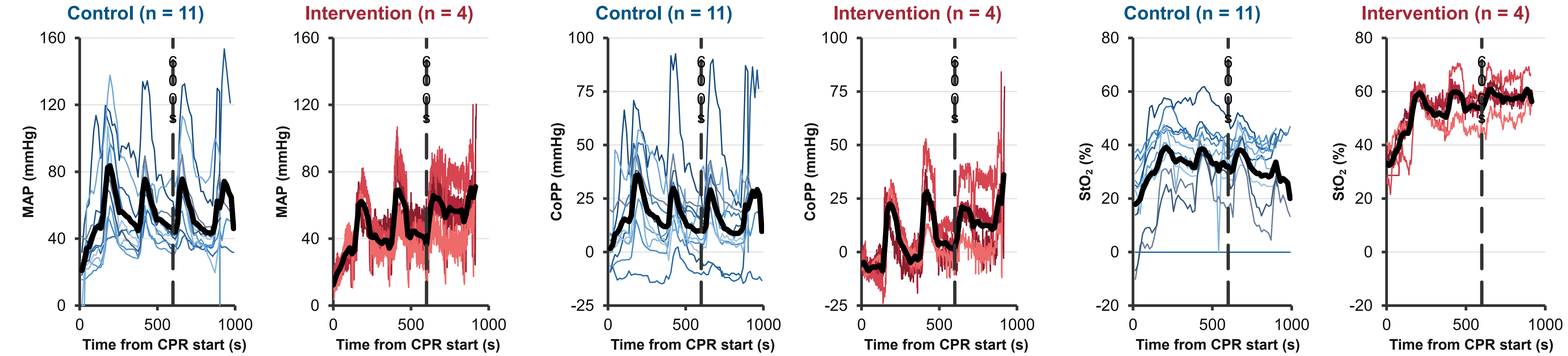



Figure 3. Individual physiologic trajectories during CPR in Control (n = 11) and Intervention (n = 4) animals; animals with massive hemorrhage. Thin lines show individual animals; the thick cyan line shows the cohort mean (equal weight per animal, 15-s bins). The vertical dashed line marks intervention deployment at 600 s. Y-axes are matched within each variable so cohorts can be compared directly. Relative cerebral blood volume (rTHC), relative cerebral blood flow (rDCS) did not demonstrate significant differences. MAP, mean arterial pressure; CoPP, coronary perfusion pressure; $StO_2$, tissue oxygen saturation.

## CONCLUSIONS

- NOM-guided dAO was feasible in an animal model of prolonged pediatric CPR.
- Compared with standard CPR, dAO increased: time within the cerebral $HbO_2$ target, systolic blood pressure and cerebral oxygenation.
- CoPP and MAP trended higher but were not significant.
- Limitations: small sample, historical (non-contemporaneous) controls, hemorrhagic complications, deflated catheter placed prior to arrest.

## FUTURE DIRECTIONS

- Perform larger, randomized study with concurrent control group.
- Assess longer-term survival and neurologic recovery with a survival cohort.
- Evaluate efficacy in additional physiologic models (hemorrhagic shock, septic shock, etc.).
- Refine balloon volume titration mechanics with the ultimate goal of **closed-loop, autonomous, and personalized resuscitation.**

## ACKNOWLEDGMENTS

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